Positive Codegree Bounds for Daisies and Graph Expansions

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Abstract

The minimum positive codegree of an r-uniform hypergraph is the minimum degree of an $(r-1)$-set contained in an edge. We prove that the positive codegree density of the four-uniform daisy is at most $6/11$, improving the upper bound $4/7$ obtained from three-uniform links. Together with the known lower bound, this gives the interval $[8/15, 6/11]$. The proof uses an exact certificate on six vertices and gives a finite bound for all orders at least 3814. Taking links yields the upper bound $6/11$ in every higher uniformity. As applications of established shadow-graph methods, we also determine the positive codegree density of a graph expansion from its chromatic number and prove edit stability. Existing partiteness theorems yield an exact finite formula for clique expansions, with uniqueness when the number of parts divides the order.

1 Introduction

All hypergraphs in this paper are finite and simple. An r-uniform hypergraph is called an r-graph. For an $(r-1)$-set $S \subseteq V(H)$, write

$$d_H(S) = \big|\{v \in V(H) \setminus S : S \cup \{v\} \in E(H)\}\big|.$$

We call S *supported* if $d_H(S) > 0$. The minimum positive codegree is

$$\delta^+_{r-1}(H) = \min\{d_H(S) : S \text{ is supported}\},$$

with value zero when H is empty. For a fixed r-graph F, define

$$\mathrm{co}^+\mathrm{ex}(n, F) = \max\{\delta^+_{r-1}(H) : |V(H)| = n,\ H \text{ is } F\text{-free}\}, \qquad \gamma^+(F) = \lim_{n\to\infty} \frac{\mathrm{co}^+\mathrm{ex}(n, F)}{n}.$$

The limit exists; see Balogh, Halfpap, Lidický, and Palmer [2] and the references there. Containment throughout means subgraph containment, without an inducedness requirement.

For $r \geq 3$, the *r-daisy* $\mathcal{D}_r$ has a set of $r-2$ stem vertices and four further vertices. Its six edges are the unions of the stem with all pairs of further vertices. Equivalently, an r-graph is $\mathcal{D}_r$-free if every $(r-2)$-set has a K_4-free graph as its link. Balogh et al. [2, Question 6.6] ask for $\gamma^+(\mathcal{D}_r)$ when $r \geq 4$. Their equality $\gamma^+(\mathcal{D}_3) = 4/7$ gives the upper bound $4/7$ in higher uniformities by taking links.

The binary-basis construction of Ellis, Ivan, and Leader [4] gives a lower bound bounded away from zero. Hou, Hou, Liu, Zeng, and Zhang [8, Theorem 1.4] explicitly obtain

$$\gamma^+(\mathcal{D}_r) \geq \frac{2^{r-1}}{2^r - 1}.$$

Our first result improves the upper bound.

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Theorem 1.1. *Every $\mathcal{D}_4$-free 4-graph H on $n \geq 3814$ vertices satisfies*

$$\delta_3^+(H) < \frac{6n}{11}.$$

Consequently,

$$\frac{8}{15} \leq \gamma^+(\mathcal{D}_4) \leq \frac{6}{11}.$$

The interval has width $2/165$, compared with $4/105$ for the interval $[8/15, 4/7]$. The exact value remains undetermined. The proof is a finite positive semidefinite certificate in the spirit of the flag algebra method [12]. We give the averaging argument explicitly, including the treatment of unsupported triples and the error from sampling distinct vertices. The accompanying integer certificate and two independent verification programs suffice to check its finite part.

Corollary 1.2. *For every fixed $r \geq 4$,*

$$\frac{2^{r-1}}{2^r - 1} \leq \gamma^+(\mathcal{D}_r) \leq \frac{6}{11}.$$

More precisely, every $\mathcal{D}_r$-free r-graph H on $n \geq 3810 + r$ vertices satisfies

$$\delta_{r-1}^+(H) < \frac{6(n - r + 4)}{11}.$$

We next consider expansions. If R is a simple graph, its *three-uniform expansion* R^+ is obtained by adding a distinct new vertex z_e to each edge $e \in E(R)$ and replacing e by the triple $e \cup \{z_e\}$. A complete m-partite 3-graph contains exactly the triples meeting three distinct parts. It is balanced if its part sizes differ by at most one. Two 3-graphs on the same vertex set differ in t triples if the symmetric difference of their edge sets has size t.

Theorem 1.3. *Let R be a fixed simple graph containing an edge, and let $s = \chi(R)$. Then*

$$\gamma^+(R^+) = \max\left\{0, \frac{s-3}{s-1}\right\}.$$

If $s \geq 4$ and (H_n) is a sequence of R^+-free 3-graphs with $|V(H_n)| = n$ and

$$\delta_2^+(H_n) \geq \left(\frac{s-3}{s-1} - o(1)\right) n,$$

then H_n differs in $o(n^3)$ triples from a complete balanced $(s-1)$-partite 3-graph.

For clique expansions, existing partiteness theorems also give a finite consequence. For integers $m \geq 3$ and $n \geq 0$, put

$$f_m(n) = n - 2\left\lfloor \frac{n}{m} \right\rfloor - \min\{2, n \bmod m\}, \qquad N_m = \left\lfloor \frac{2m(m-1)(3m-1)}{m-2} \right\rfloor + 1. \tag{1}$$

Corollary 1.4. *For every integer $m \geq 3$ and every $n \geq N_m$,*

$$\mathrm{co}^+\mathrm{ex}(n, K_{m+1}^+) = f_m(n).$$

If $m \mid n$, the unique extremal 3-graph, up to isomorphism, is the complete balanced m-partite 3-graph.

The expansion results use established shadow-graph arguments. Hou, Li, Yang, Zeng, and Zhang [9] combine shadows, clique inequalities, and graph degree theorems in their work on saturated Berge-clique-free hypergraphs. For $m \geq 4$, Corollary 1.4 is an explicit consequence of their minimum-degree partiteness theorem [9, arXiv version, Theorem 5.2], together with the shadow reduction and link-degree estimate below. Liu, Ren, and Wang [10, Theorem 3.2] prove a stronger partiteness theorem covering $m = 3$. We include a uniform proof through the Andrásfai–Erdős–Sós graph theorem, followed by the integer calculation. The finite formula does not resolve the sharper saturation-removal problem of [9].

Theorem 1.3 also relates to the attainable-value part of [2, Question 6.5]. The clique family gives the values $(m-2)/m$. Some of these values were already known for other forbidden hypergraphs: $3/5$ occurs in [2], and Bodnár, Gao, Pikhurko, Rong, and Sun [3, Corollary 4] realize $(a-1)/a$ for every positive integer a, covering every even m here. Our statements determine the densities of the specified expansions and give their finite and stability consequences. They do not classify all attainable values or determine the jumps of γ^+.

Section 2 proves the daisy bound. Sections 3 and 4 prove the asymptotic and finite expansion results, respectively. We write $e(H) = |E(H)|$, and $k_j(G)$ for the number of j-vertex cliques in a graph G.

2 A six-vertex certificate for daisies

We first record an elementary edge-counting consequence of positive codegree. It ensures that a hypergraph with positive codegree proportional to its order cannot have vanishing edge density.

Lemma 2.1. *If H is a nonempty r-graph and $\delta = \delta^+_{r-1}(H)$, then*

$$r!e(H) \geq \delta^r.$$

Proof. Fix an ordered edge $(v_1, \dots, v_r)$. Replace its first coordinate by a neighbour of the other $r-1$ coordinates, then replace the second coordinate, and continue through all r coordinates. At every step the current tuple is an ordered edge, so there are at least δ choices. There are therefore at least δ^r replacement sequences. The final tuple determines each chosen replacement, because every coordinate is replaced exactly once. Thus these sequences have distinct final ordered edges, and there are $r!e(H)$ ordered edges in total. □

We now define the local quantities used in the certificate. Put $q = 6/11$. For a 4-graph F on six vertices, choose an ordered list (a, b, c, d, x, y) of its vertices uniformly. Write $U = (U_0, U_1, U_2, U_3) = (a, b, c, d)$, and set

$$t = \mathbf{1}_{\{a,b,c,d\} \in E(F)}, \qquad f_x = \sum_{j=0}^{3} 2^j \mathbf{1}_{(\{a,b,c,d\} \setminus \{U_j\}) \cup \{x\} \in E(F)}.$$

Define f_y in the same way. For $1 \leq f \leq 15$, let $u_0(f)$ be the fth coordinate vector of $\mathbb{R}^{15}$, and put $u_0(0) = 0$. For $0 \leq f \leq 15$, let $u_1(f)$ be the $(f+1)$st coordinate vector of $\mathbb{R}^{16}$.

The certificate contains integer matrices R_0 and R_1 of sizes 15×15 and 16×16. With $s_0 = 10^6$, define

$$Q_0 = \frac{R_0 R_0^{\mathsf{T}}}{s_0^2}, \qquad Q_1 = \frac{R_1 R_1^{\mathsf{T}}}{s_0^2}, \qquad S(F) = \mathbb{E}\big[u_t(f_x)^{\mathsf{T}} Q_t u_t(f_y)\big].$$

Both matrices are positive semidefinite over the reals. Omitting the zero extension for a nonedge root means assigning it the zero vector, so no conditioning on the presence of an extension is involved.

For the positive codegree constraints, choose a uniform ordering (a, b, c, x, y, z) and form the five-bit flag

$$f = \mathbf{1}_{abcx\in E(F)} + 2\mathbf{1}_{abcy\in E(F)} + 4\mathbf{1}_{abxy\in E(F)} + 8\mathbf{1}_{acxy\in E(F)} + 16\mathbf{1}_{bcxy\in E(F)}.$$

Here and below a string of distinct vertices denotes its underlying set. Let $B_f(F)$ be the probability of obtaining this flag, and let $A_f(F)$ be the probability of obtaining the flag and also having $abcz \in E(F)$. We use only the set

$$\mathcal{I} = \{f \in \{0, \dots, 31\} : f \bmod 4 \neq 0\}.$$

Every flag in $\mathcal{I}$ certifies that abc is supported. For each $f \in \mathcal{I}$, the certificate gives a nonnegative integer ℓ_f; put $\lambda_f = \ell_f/s_0$. Finally, write $d_4(F) = e(F)/15$.

Lemma 2.2 (Exact local certificate)**.** *The matrices and multipliers in the accompanying file* `daisy_6_11_certificate.json` *satisfy, for every $\mathcal{D}_4$-free 4-graph F on six vertices,*

$$S(F) + \sum_{f\in\mathcal{I}} \lambda_f(A_f(F) - qB_f(F)) \leq -d_4(F). \tag{2}$$

Moreover, the largest diagonal entry of Q_0 and Q_1 is

$$M = \frac{16856850131813}{50000000000}. \tag{3}$$

Proof. This is the finite, computer-assisted part of the proof. We describe the full exact check. Encode a labelled 4-graph on $\{0, 1, \dots, 5\}$ by the 15 possible edges in lexicographic order. Enumerate the 2^{15} edge sets and discard an edge set if, for some pair of stem vertices, all six unions of that pair with pairs of the remaining four vertices occur. This leaves 27626 labelled graphs in 120 isomorphism classes. Vertex permutations give the orbits, so checking one representative per orbit covers every legal edge set.

For each representative, average the expressions defining S, A_f, and B_f over the 720 vertex orders. Multiplication by $720 \cdot 11 \cdot s_0^2$ gives an integer. A Gram contribution is 11 times an entry of $R_t R_t^{\mathsf{T}}$, while a codegree contribution is

$$s_0\ell_f(11\mathbf{1}_{abcz\in E(F)} - 6).$$

The verifier adds these integers and compares their sum, multiplied by 15, with $-720 \cdot 11 \cdot s_0^2 e(F)$. The empty graph gives zero. On the other 119 types, the maximum of the left side of (2) divided by $d_4(F)$ is

$$-\frac{2000931390921}{2000000000000} < -1.$$

It also checks that all entries of the Gram factors are integers, that their dimensions are as stated, and that the 24 multipliers are nonnegative. Computing all diagonal Gram entries gives (3).

The file `verify_daisy_certificate.py` performs these operations using only integer arithmetic and exact fractions. A second program, `independent_daisy_audit.py`, represents each four-edge by its complementary pair on the six vertices. In this representation the daisy test becomes an ordinary K_4 test. It averages over unordered roots and then symmetrizes the root orders, independently recovering all 120 coefficients. Neither implementation uses a numerical optimization output or a stored coefficient table. □

Proof of Theorem 1.1. Suppose that H is $\mathcal{D}_4$-free on $n \geq 3814$ vertices and that $\delta_3^+(H) \geq qn$. In particular, H is nonempty. Define $S(H)$, $A_f(H)$, and $B_f(H)$ by sampling ordered distinct vertices from H using the same formulas as above. Apply Lemma 2.2 to every induced six-vertex subgraph. Averaging (2) gives

$$S(H) + \sum_{f\in\mathcal{I}} \lambda_f(A_f(H) - qB_f(H)) \leq -d_4(H), \qquad d_4(H) = \frac{e(H)}{\binom{n}{4}}. \tag{4}$$

The edge-density term has this normalization because every four-set is sampled equally often.

Condition on an ordered five-tuple (a, b, c, x, y) with flag $f \in \mathcal{I}$. The triple abc is supported. If z is chosen uniformly outside this five-tuple, then

$$\mathbb{P}(abcz \in E(H) \mid a, b, c, x, y) = \frac{d_H(abc) - \mathbf{1}_{abcx \in E(H)} - \mathbf{1}_{abcy \in E(H)}}{n-5} \geq \frac{qn-2}{n-5} \geq q,$$

where the final inequality uses $5q - 2 = 8/11 > 0$. Hence $A_f(H) - qB_f(H) \geq 0$ for every used flag. This step does not impose a degree condition on an unsupported triple.

It remains to control $S(H)$ when its two outside vertices must be distinct. Fix an ordered four-tuple U and its type t. Put $N = n - 4$, and associate to each vertex $v \notin U$ the vector $w_v = u_t(f_v)$. Positive semidefiniteness gives

$$\begin{aligned}\mathbb{E}_{x \neq y}[w_x^{\mathsf{T}} Q_t w_y] &= \frac{(\sum_v w_v)^{\mathsf{T}} Q_t (\sum_v w_v) - \sum_v w_v^{\mathsf{T}} Q_t w_v}{N(N-1)} \\ &\geq -\frac{M}{N-1}.\end{aligned}$$

Indeed, each w_v is a coordinate vector or zero, so each diagonal term is at most M. Averaging over U yields $S(H) \geq -M/(n-5)$. Since the remaining terms on the left side of (4) are nonnegative, we obtain

$$d_4(H) \leq \frac{M}{n-5}.$$

On the other hand, Lemma 2.1 implies

$$d_4(H) \geq \frac{\delta_3^+(H)^4}{n(n-1)(n-2)(n-3)} \geq q^4.$$

Together these inequalities force $n \leq 5 + M/q^4$. The exact value is

$$5 + \frac{M}{q^4} = \frac{247125142779874133}{64800000000000} < 3814,$$

contrary to our assumption. This proves the finite upper bound. The lower bound is the case $r = 4$ of the known construction recalled below. □

For completeness, the binary-basis construction has one vertex type for each nonzero vector of $\mathbb{F}_2^r$. An r-set of types is an edge exactly when it is a basis. Replace each type by a class of equal size and take all corresponding transversal edges. Every supported $(r-1)$-set has 2^{r-1} completing types out of $2^r - 1$. A daisy cannot occur: after quotienting by the span of its stem, its four petal types would determine four distinct nonzero directions in a two-dimensional binary space, which has only three. Repeated types cannot occur together in a basis edge. This gives the lower bound stated in [8, Theorem 1.4].

Proof of Corollary 1.2. The lower bound is the construction just described. For the upper bound, we may assume that H is nonempty. Choose an $(r-4)$-set A contained in an edge, allowing $A = \varnothing$ when $r = 4$. Its link $G = L_H(A)$ is a nonempty 4-graph on $n - r + 4$ vertices. A copy of $\mathcal{D}_4$ in G, together with A, would give a copy of $\mathcal{D}_r$ in H. Moreover, for each supported triple T of G,

$$d_G(T) = d_H(A \cup T) \geq \delta_{r-1}^+(H).$$

Theorem 1.1, applied to G, proves the finite inequality. Dividing by n and taking the limit proves the density bound. □

3 Graph expansions and stability

For a 3-graph H, its *shadow* ∂H is the graph on $V(H)$ whose edges are the supported pairs. Every edge of H is a triangle in ∂H. We use two standard observations about shadows.

Lemma 3.1. *Let R be a graph on v vertices with t edges. If H is an R^+-free 3-graph and $\delta_2^+(H) > v + t$, then ∂H is R-free. In the special case $R = K_{m+1}$, it is enough that*

$$\delta_2^+(H) > m + \binom{m+1}{2} - 2.$$

Proof. Suppose a copy of R lies in ∂H. Choose the private vertices for its expansion one edge at a time. When j private vertices have been chosen, a core pair has at most $v - 2 + j$ forbidden neighbours: the other core vertices and the previously chosen private vertices. Since $j \le t - 1$, positive codegree greater than $v + t - 3$ suffices for every choice. This proves both stated bounds. □

Lemma 3.2. *If J is a K_{m+1}-free graph on n vertices, where $m \ge 3$, then*

$$3k_3(J) \le e(J)(m-2)\sqrt{\frac{2e(J)}{m(m-1)}} \le \frac{m-2}{m}\, ne(J). \tag{5}$$

Proof. The clique-number form of Maclaurin's inequality [5, 11] gives

$$k_3(J) \le \binom{m}{3}\left(\frac{e(J)}{\binom{m}{2}}\right)^{3/2}.$$

For clique number less than m, this follows by applying the inequality with the actual clique number and using that $(m-2)/\sqrt{m(m-1)}$ is increasing for $m \ge 2$. The triangle-free case is immediate. Rearranging gives the first inequality in (5); Turán's bound $e(J) \le (m-1)n^2/(2m)$ gives the second. □

Proof of Theorem 1.3. We first establish a reduction for a sequence of R^+-free 3-graphs H_n with $\delta_2^+(H_n) \ge dn$, where $d > 0$ is fixed. Write $G_n = \partial H_n$. Lemma 3.1 implies that G_n is R-free for sufficiently large n. Since R is s-colourable, a fixed balanced complete s-partite graph contains R when its parts are large enough. The standard complete-partite supersaturation consequence of the Erdős–Stone theorem therefore gives $k_s(G_n) = o(n^s)$. The graph removal lemma [6] allows us to delete $o(n^2)$ edges and obtain a K_s-free graph J_n.

We will divide by $e(G_n)$, so we first check that it has order n^2. Choose a vertex v contained in an edge of H_n. Its link, on the neighbourhood of v in G_n, has no isolated vertices and has minimum degree at least dn. It has at least dn vertices, and hence at least $d^2n^2/2$ edges. All these edges belong to G_n. Thus

$$e(G_n) = \Omega(n^2), \qquad e(J_n) = e(G_n) - o(n^2). \tag{6}$$

The deletions destroy at most $o(n^3)$ triangles. Summing the pair codegrees gives

$$\delta_2^+(H_n)e(G_n) \le 3e(H_n) \le 3k_3(G_n) = 3k_3(J_n) + o(n^3). \tag{7}$$

If $s \le 3$, then J_n is triangle-free, so (7) contradicts (6) and $\delta_2^+(H_n) \ge dn$. No sequence with positive limiting codegree exists in this case, proving $\gamma^+(R^+) = 0$.

Suppose now that $s \ge 4$, and put $m = s - 1$ and $c = (m-2)/m$. Lemma 3.2 and (7) give

$$\delta_2^+(H_n)e(G_n) \le cne(J_n) + o(n^3) \le cne(G_n) + o(n^3).$$

Division using (6) proves $\delta_2^+(H_n) \le cn + o(n)$. This excludes every subsequence with limiting codegree greater than c.

For the matching lower bound, take a complete balanced m-partite 3-graph. A copy of R^+ would give a proper m-colouring of the core graph R, by assigning each core vertex its part. This is impossible because $\chi(R) = m + 1$. The construction has minimum positive codegree $(c + o(1))n$, proving the density formula.

It remains to prove stability. Assume $\delta_2^+(H_n) \ge (c - o(1))n$. Since $c > 0$, the preceding reduction applies with $d = c/2$ for large n. The first inequality of (5), together with (7), gives

$$\begin{aligned}(c - o(1))n &\le \frac{e(J_n)}{e(G_n)}(m-2)\sqrt{\frac{2e(J_n)}{m(m-1)}} + o(n)\\ &\le (m-2)\sqrt{\frac{2e(J_n)}{m(m-1)}} + o(n).\end{aligned}$$

Consequently,

$$e(J_n) \ge \frac{m-1}{2m}n^2 - o(n^2).$$

Turán's theorem supplies the reverse asymptotic inequality. The graph stability theorem, in the form of [7], gives an asymptotically balanced m-partition for which J_n differs in $o(n^2)$ pairs from the complete multipartite graph T_n. The same holds for G_n, because only $o(n^2)$ edges were deleted.

The triangle counts of G_n and T_n differ by $o(n^3)$. On the other hand,

$$e(H_n) \ge \frac{\delta_2^+(H_n)e(G_n)}{3} \ge \binom{m}{3}\frac{n^3}{m^3} - o(n^3).$$

The main term on the right is also the triangle count of T_n. Since every edge of H_n is a triangle of G_n, the hypergraph H_n omits only $o(n^3)$ triangles of G_n. It therefore differs in $o(n^3)$ triples from the complete m-partite 3-graph associated with T_n. Moving $o(n)$ vertices makes the part sizes differ by at most one, at a further cost of $o(n^3)$ triple changes. This proves the required stability. □

4 The finite value for clique expansions

We prove Corollary 1.4 by passing from large positive codegree to a multipartite shadow. The graph theorem we need is due to Andrásfai, Erdős, and Sós [1]: every K_{m+1}-free graph on N vertices with minimum degree greater than $(3m-4)N/(3m-1)$ is m-partite.

Lemma 4.1. *Let H be a 3-graph whose shadow G is K_{m+1}-free, where $m \ge 3$. For every nonisolated vertex v of G,*

$$d_G(v) \ge \frac{m-1}{m-2}\delta_2^+(H).$$

Proof. The link $L_H(v)$, with vertex set $N_G(v)$, has no isolated vertices. Each of its vertices u has degree $d_H(uv) \ge \delta_2^+(H)$. Moreover, it is K_m-free: a clique of order m in the link, together with v, would give a clique of order $m+1$ in G. Turán's theorem therefore yields

$$\delta_2^+(H) \le \frac{2e(L_H(v))}{d_G(v)} \le \frac{m-2}{m-1}d_G(v).$$

□

Proof of Corollary 1.4. Fix $m \geq 3$ and $n \geq N_m$. In a complete balanced m-partite 3-graph, the codegree of a pair in different parts is n minus the sum of their part sizes. The two largest parts have total size

$$2\left\lfloor \frac{n}{m} \right\rfloor + \min\{2, n \bmod m\}.$$

Thus the minimum positive codegree is $f_m(n)$, as defined in (1). The graph is K^+_{m+1}-free, since the $m+1$ core vertices of a putative expansion would have to lie in distinct parts. This proves the lower bound.

Let H be extremal, and abbreviate

$$\delta = \delta_2^+(H), \qquad c = \frac{m-2}{m}, \qquad a = \frac{m-1}{m-2}, \qquad h = \frac{3m-4}{3m-1}.$$

The construction gives $\delta \geq f_m(n) \geq cn - 2 > 0$. The choice of N_m means that $n > 2am(3m-1)$. In particular,

$$cn - 2 > 2(m-1)(3m-1) - 2 > m + \binom{m+1}{2} - 2,$$

where the second inequality follows from $11m^2 - 19m + 4 > 0$ for $m \geq 3$. Lemma 3.1 shows that the shadow $G = \partial H$ is K_{m+1}-free.

Delete the isolated vertices of G, leaving a graph G_0 on $N \leq n$ vertices. Lemma 4.1 gives

$$\delta(G_0) \geq a\delta \geq \frac{m-1}{m} n - 2a > hn \geq hN. \tag{8}$$

Here the strict inequality uses

$$\frac{m-1}{m} - h = \frac{1}{m(3m-1)}, \qquad n > 2am(3m-1).$$

The Andrásfai–Erdős–Sós theorem partitions G_0 into at most m independent sets. Pad with empty sets if necessary and order their sizes as $b_1 \geq \cdots \geq b_m \geq 0$.

There is an edge of G_0 between its two largest classes. Otherwise every vertex of the largest class has degree at most

$$N - b_1 - b_2 \leq \frac{m-2}{m} N \leq cn,$$

whereas (8) gives degree at least $(m-1)n/m - 2a > cn$, because $n > 2am$. Choose a supported pair joining these two classes. Its neighbours in H lie outside both classes. Hence

$$\delta \leq N - b_1 - b_2 \leq f_m(N) \leq f_m(n). \tag{9}$$

We justify the two integer comparisons explicitly. Write $N = mk + r$, where $0 \leq r < m$. If $r = 0$ and $b_1 + b_2 \leq 2k - 1$, then $b_2 \leq k - 1$ and

$$N \leq 2k - 1 + (m-2)(k-1) < mk.$$

If $r = 1$ and $b_1 + b_2 \leq 2k$, then $b_2 \leq k$ and $N \leq mk$. If $r \geq 2$ and $b_1 + b_2 \leq 2k + 1$, then again $b_2 \leq k$ and $N \leq mk + 1$. Each case contradicts $N = mk + r$. Thus $b_1 + b_2 \geq 2k + \min\{2, r\}$, proving the middle inequality in (9). Finally, $f_m(n+1) - f_m(n)$ is zero when $n \bmod m$ is 0 or 1, and is one otherwise. Thus f_m is nondecreasing. The lower construction and (9) prove the exact value.

Suppose now that $m \mid n$, put $k = n/m$, and assume $\delta = (m-2)k = cn$. The preceding partition satisfies

$$cn \leq N - b_1 - b_2 \leq cN \leq cn.$$

All inequalities are equalities. Hence $N = n$ and $b_1 + b_2 = 2k$. We have $b_2 \le k$, while

$$mk = \sum_{i=1}^{m} b_i \le b_1 + (m-1)b_2 = 2k + (m-2)b_2$$

forces $b_2 \ge k$. Therefore every class has size k.

Each supported pair has at most $n - 2k = \delta$ possible neighbours, and thus has every vertex outside its two classes as a neighbour. Starting with any edge, one may consequently replace a vertex by any vertex outside the other two classes and still obtain an edge. The three-element subsets of the set of classes are connected by replacements of one class at a time. After reaching any desired three classes, replace the individual vertices within those classes. Every triple in three distinct classes is reached in this way. Thus H is the complete balanced m-partite 3-graph, proving uniqueness. □

For example, $N_7 = 337$, so $\mathrm{co}^+\mathrm{ex}(n, K_8^+) = n - 2\lfloor n/7 \rfloor - \min\{2, n \bmod 7\}$ for every $n \ge 337$, and $\gamma^+(K_8^+) = 5/7$. The sufficient order N_m was chosen to keep the argument uniform; no optimality is asserted for it. The uniqueness statement concerns $m \mid n$ only.

Computational material and data availability

The accompanying `supplement` directory contains the integer certificate, the two exact verifiers, a script running all checks, and a description of their encodings. The certificate has SHA-256 digest

`9ee628bede8d4d5c9f8426cdafb464c71ab`
`0285ae83d5a541737ee0fa8bd93f9`.

GPT-6.1 (OpenAI) assisted with implementing the verification programs. All certificate verification uses the Python standard library. The expansion proofs are analytic; the additional finite checks supplied for them are consistency checks, not substitutes for the proofs.

Declaration of AI use

The author used GPT-6.1 to assist with exploring proof strategies, checking calculations and proofs, and improving the exposition. The author takes full responsibility for the content of this paper.

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